Skeletal-Feature Adaptive Visual Servoing: An Occlusion-Robust Vision-Control Framework for Robotic Manipulators

Md Hasibuzzaman[1,*] and Md Shetu Mia[2]

Abstract

Image-based visual servoing drives a robot toward a target using features tracked directly in the camera image, and the choice of feature matters more than most controllers admit. Centroid features are cheap but blind to rotation; corner or hull features carry full pose information but disappear or jump the moment a hand, cable, or passing object occludes them. This paper introduces SF-AIBVS, a visual servoing controller built on the target's morphological skeleton rather than its raw silhouette. Because a skeleton branch degrades in length as its corresponding part of the object is covered, rather than vanishing outright, the resulting position, scale, and orientation estimates stay usable well past the point where corner detectors lose track entirely. We pair this with a confidence gate: when the observed skeleton length falls far short of what the current scale estimate predicts, the controller freezes its output instead of acting on a corrupted reading, and resumes the instant the reading is trustworthy again. Across 360 simulated docking trials at three occlusion severities, the skeletal controller reached the target in all 90 trials; a classical centroid controller reached it in none, and a corner-based tracker reached it in 13 to 20 trials, diverging outright in up to 77% of heavy-occlusion attempts. Disabling the confidence gate does not change the success count, but it does make the control signal visibly rougher under heavy occlusion (a 19% increase in mean command-to-command variation, $p = 0.007$), so the gate's contribution is best described as a smoothness and safety property layered on top of an already robust feature choice, not a rescue mechanism for a fragile one. A sweep of the confidence threshold from 0.30 to 0.90 shows performance is stable across a wide plateau rather than dependent on a narrowly tuned value. All of this is evaluated in simulation on a single target shape with a reduced, four-parameter pose model and idealized segmentation, and we treat that scope, along with the corresponding need for real-camera validation, as the central limitation of the work rather than an afterthought.

Keywords

image-based visual servoing, occlusion robustness, skeletonization, medial axis, uncertainty-gated control, intelligent automation, AI-driven vision

I. Introduction

A robot arm reaching for a tool it can partly see is a more common situation than most visual servoing papers let on. Warehouse picking scenes have overlapping boxes; surgical and inspection cameras work through smoke, fluid, or other instruments; a household robot's target mug is routinely half-hidden behind a hand or a second mug. Image-based visual servoing (IBVS) handles the well-lit, unoccluded case well, but the moment the visual feature set it depends on gets partially blocked, most classical schemes either lose the feature outright or silently absorb a biased estimate into the control loop.

The feature choice is the crux of the problem, more so than the control law built on top of it. A centroid, computed as the mean pixel position of the visible silhouette, is trivial to extract and remarkably stable under noise, but it carries no information about the object's rotation, and under asymmetric occlusion its estimate of position shifts toward whichever part of the object remains visible. Corner or hull-based schemes recover full six-parameter pose information and have decades of practical use behind them [1], [2], but they depend on a fixed set

[1]M. Hasibuzzaman is with the Department of Computer Science and Information Engineering, Asia University, No. 500 Liufeng Road, Wufeng District, Taichung 41354, Taiwan.

[2]M. S. Mia is with the Department of Physics, Astronomy and Materials Science, Missouri State University, 901 S. National Avenue, Springfield, MO 65897, USA.

[*]Corresponding author. E-mail: 113121017@live.asia.edu.tw.

of point correspondences that a real detector must re-find every frame; when a corner is covered, the detector either drops it, causing a discontinuity in the fitted pose, or worse, matches a different, incorrect point to it. Section VI reproduces both failure modes directly.

This paper explores a third option: use the target's morphological skeleton, the thin, one-pixel-wide medial axis of its silhouette [3], as the servoing feature itself. The underlying idea of tracking a quantity's reliability alongside the quantity itself is a familiar one in robotics state estimation more broadly [4], though it is rarely applied to the feature layer of a visual servoing loop specifically. Skeletonization is a mature, standard operation in image processing [5] but has, to our knowledge, not previously been used as the feature representation driving the control law in an IBVS loop. The property we exploit is structural: occluding one arm of a branching shape removes the corresponding skeleton branch but leaves the rest of the skeleton graph, including its junction and remaining endpoints, intact. Where a corner detector loses a discrete point entirely, a skeleton loses a fraction of its length, degrading its estimate gracefully rather than catastrophically.

We combine this feature with a lightweight confidence gate. Because we track how much skeleton length we would expect to see given the current scale estimate, a sudden shortfall is a direct, interpretable signal that occlusion has become severe enough that this frame's reading should not be trusted. Rather than build a predictive model to paper over the gap, as we describe in Section IV, we simply hold the previous command until the reading recovers. This choice was not the first one we tried; an earlier version of the gate extrapolated feature velocity through the occlusion and turned out to accumulate drift indefinitely once the target stopped moving in the way the extrapolation assumed. We keep that failure in the discussion (Section VII) because it shaped the final, simpler design and because the same failure mode is easy to reintroduce by accident in related systems.

I-A. *Contributions*

- A skeleton-based feature set for image-based visual servoing, together with an unambiguous, full-range orientation estimate derived from the skeleton's endpoint structure rather than its principal axis (a PCA-based axis is only defined modulo 180 degrees, which is insufficient for a target that is not symmetric under half-turn rotation).
- A confidence gate driven by the ratio of observed to expected skeleton length, which freezes the control output during severe occlusion and resumes it automatically once the target is visible enough again.
- A controlled comparison against centroid and corner-based IBVS across 360 simulated trials spanning three occlusion severities, with an ablation isolating exactly what the confidence gate does and does not contribute, reported without inflating the gate's role beyond what the data show.

I-B. *Paper Organization*

Section II places this work relative to classical IBVS, occlusion-robust tracking, and skeleton-based shape description. Section III defines the servoing task. Section IV details SF-AIBVS. Section V describes the simulated environment, baselines, and evaluation protocol. Section VI reports outcomes. Section VII discusses what the confidence gate actually buys, and what it does not. Section VIII concludes.

II. Related Work

II-A. *Image-Based Visual Servoing*

Sanderson and Weiss gave the field its still-standard vocabulary, splitting visual servo control into position-based schemes, which reconstruct a 3D pose before controlling it, and image-based schemes, which control image-space error directly [6]. Espiau, Chaumette, and Rives later formalized the image-based case as a control problem defined directly in feature space rather than Cartesian space [1], and Hutchinson, Hager, and Corke's tutorial [2], extended by Chaumette and Hutchinson in two parts [7], [8], remains the standard reference for the field's basic and advanced control formulations. All of these treatments are, by design, feature-agnostic: they describe how to build a control law once an interaction matrix relating feature velocity to camera velocity is known, and they leave the choice of feature, points, lines, image moments, or otherwise, to the application. Point features (corners, markers, or blob centroids) dominate practical deployments because they are cheap to extract and their interaction matrices are well understood in closed form. Our contribution sits entirely within this gap: we propose a different feature, not a different control law, and inherit the same proportional control structure used throughout the classical literature.

II-B. Occlusion-Robust Tracking

The general vision problem of tracking an object through occlusion has its own long history, most influentially through kernel-based and mean-shift trackers [9], which represent the target as a color or intensity histogram and localize it by gradient ascent on a similarity score. These methods are robust to partial occlusion in the sense that a histogram computed over a partly-visible region still resembles the full-object histogram, but they were designed for 2D image-plane tracking (bounding box localization), not for recovering the rotation and scale parameters an IBVS control law needs. Extending a histogram-based tracker to output a full similarity transform is possible but not standard practice; skeleton features, by contrast, yield position, scale, and orientation from the same graph structure without additional machinery.

II-C. Skeletonization and Medial Axis Representations

The medial axis transform [3] and its discrete thinning approximations [5] are foundational tools in shape analysis, optical character recognition, and biomedical image processing, prized for reducing a two-dimensional silhouette to a compact, topologically meaningful one-dimensional graph. Their use has been almost entirely off-line and descriptive: computing a skeleton to measure, classify, or compress a shape, rather than to close a real-time feedback loop. We are not aware of prior work that places the skeleton graph inside the servoing loop itself as the feature vector a controller acts on every frame; this paper's method section describes the specific adaptations, an unambiguous orientation estimate and a length-ratio confidence signal, needed to make that practical.

II-D. Learned and Predictive Alternatives

Deep learning has also been applied directly inside the servo loop: Bateux et al. train a convolutional network to regress the relative pose between current and desired images and report robustness to occlusion and lighting change as an emergent property of a large synthetic training set [10], and decentralized collision-avoidance policies show a related capacity to implicitly anticipate the motion of unmodeled agents [11]. Such approaches can be highly capable but require training data representative of the deployment distribution and are difficult to certify against a specific, stated failure mode, in contrast to the explicit, inspectable length-ratio threshold used here. We position SF-AIBVS as a lightweight, training-free alternative suited to settings where auditability matters as much as raw performance, which is the emphasis of this special issue's intelligent control theme.

III. Problem Formulation

We consider a camera-target docking task reduced to its essential image-plane parameters: a target's appearance in the image is fully described by a translation (t_x, t_y), an isotropic scale s standing in for depth, and an in-plane rotation θ. The task is to drive (t_x, t_y, s, θ) from an unknown initial offset to a centered, canonically scaled, upright configuration, using only a binary silhouette mask of the target observed at each time step. A moving, opaque occluder may cover part of the target at any time; the controller has no access to the occluder's position, size, or motion, only to whatever fraction of the target silhouette remains visible in the current frame.

This formulation deliberately keeps the camera model orthographic and the object planar, which lets us isolate the feature-extraction and confidence-gating questions from the additional complexity of a full projective, six-degree-of-freedom interaction matrix. Section VII returns to this simplification as a limitation rather than treating it as free of consequence.

IV. Proposed Method: SF-AIBVS

IV-A. Overview

At every control step, SF-AIBVS skeletonizes the visible binary mask, reads off a position, scale, and orientation estimate from that skeleton, checks whether the reading is consistent with what recent, trusted readings would predict, and either acts on the reading or holds still. Fig. 3 lays out this sequence; Algorithm 1 gives the corresponding pseudocode.

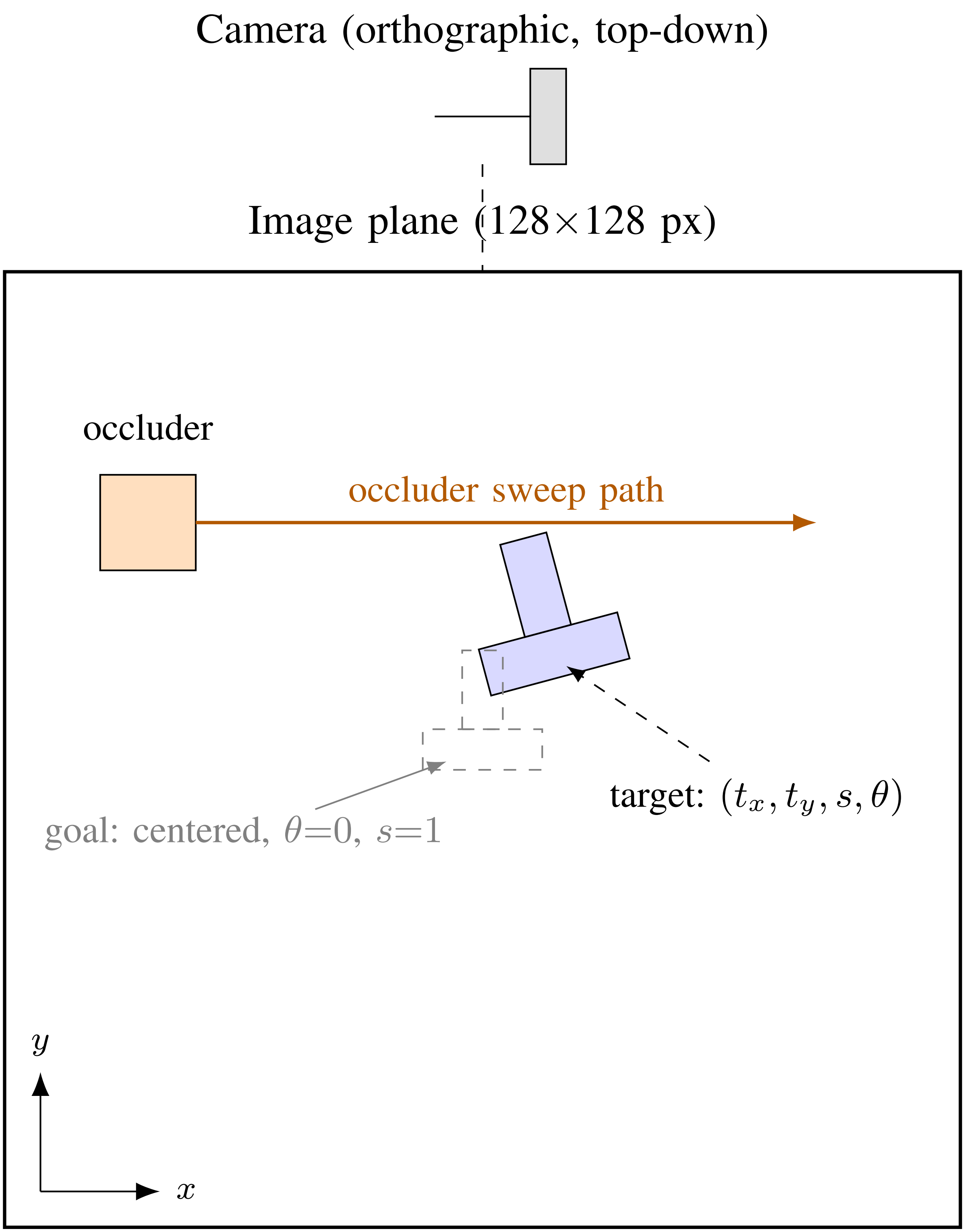

Fig. 1: The simulated docking environment. The camera observes a planar workspace from directly above; the target's pose relative to the goal is described by four parameters (t_x, t_y, s, θ); a rectangular occluder sweeps horizontally at a fixed row, independent of the target's own motion.

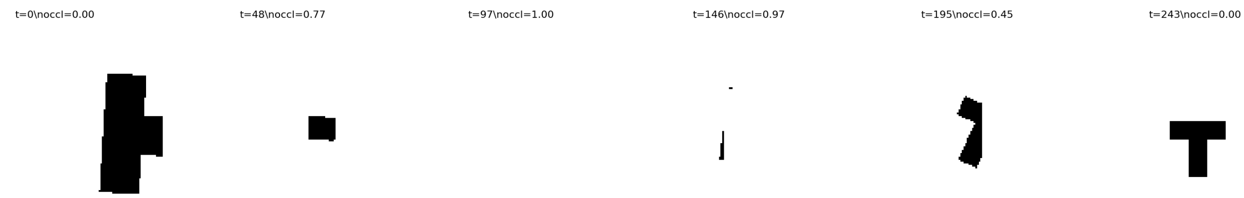

Fig. 2: A single trial from start to convergence (heavy occlusion, seed 4). The target is fully hidden at t=98 and t=146 (occlusion fraction 1.0) yet the controller recovers and reaches the goal by t=244; the occlusion column headers report the true occlusion fraction at that instant, not anything estimated by the controller.

IV-B. Feature Extraction

Given the visible mask, we compute its skeleton with a standard thinning operation [5]. Three quantities are read off this skeleton:

Position. The mean coordinate of all skeleton pixels, corrected by a fixed offset representing how far the canonical shape's own skeleton centroid sits from its nominal reference point. Without this correction, the controller would drive the skeleton centroid to the image center, which is not the same thing as centering the object; Fig. 4 shows a raw silhouette centroid suffering exactly this kind of bias as occlusion increases, and the skeleton centroid needs the same nominal correction even though it is less severely displaced.

Scale. The observed skeleton pixel count divided by the skeleton length of the canonical shape at unit scale, measured once offline.

Orientation. Rather than take the principal axis of the skeleton pixel distribution, which is a line, not a vector, and therefore leaves a 180-degree ambiguity no proportional controller can resolve on its own, we find the skeleton endpoint farthest from the skeleton's centroid and take the bearing to it (Fig. 5). For a shape with one long, distinctive branch, this endpoint reliably identifies that branch's tip, and the bearing to it is meaningful over the full rotation range rather than only half of it. This is a deliberately simple stand-in for the point-correspondence machinery behind classical keypoint descriptors such as Harris corners [12], Shi-Tomasi features [13], SIFT [14], SURF [15], and ORB [16]; those methods solve a harder problem (matching many points across arbitrary viewpoint change) and were not built with occlusion-graceful degradation as a design goal, which is precisely the property Fig. 4 is meant to demonstrate for the skeleton in contrast to a small, fixed corner set.

IV-C. Confidence Gate

A running estimate of the target's true scale, updated only when the gate judges the current frame trustworthy, predicts how many skeleton pixels should be visible this frame. The ratio of the pixels actually observed to this expectation is the confidence score. When it drops below a fixed threshold, or when the skeleton is too small to extract endpoints from at all, the controller issues a zero command for that step rather than computing one from the degraded reading, and leaves its internal scale estimate untouched. Because nothing moves while frozen, the expectation does not go stale relative to a target that has not changed, so confidence recovers cleanly the moment the occluder clears, without any separate re-acquisition logic.

This is a deliberately conservative design, and Section VII explains why: an earlier version of this gate attempted to bridge occlusion by extrapolating the feature's recent velocity forward, in the spirit of the predict step of a recursive filter [17], on the reasoning that a brief occlusion should not need to stop the controller outright. That approach worked for short gaps but produced runaway drift during long ones, because a constant extrapolated correction, applied every step against a target whose true position was no longer changing, accumulates without bound. Freezing avoids this failure mode by construction, at the cost of making no progress during occlusion rather than continuing to (possibly wrongly) close in on the goal.

IV-D. Algorithm Summary

The complete Python implementation, simulator, both baselines, SF-AIBVS, and every script used to produce the tables and figures in this paper, is provided as supplementary code.

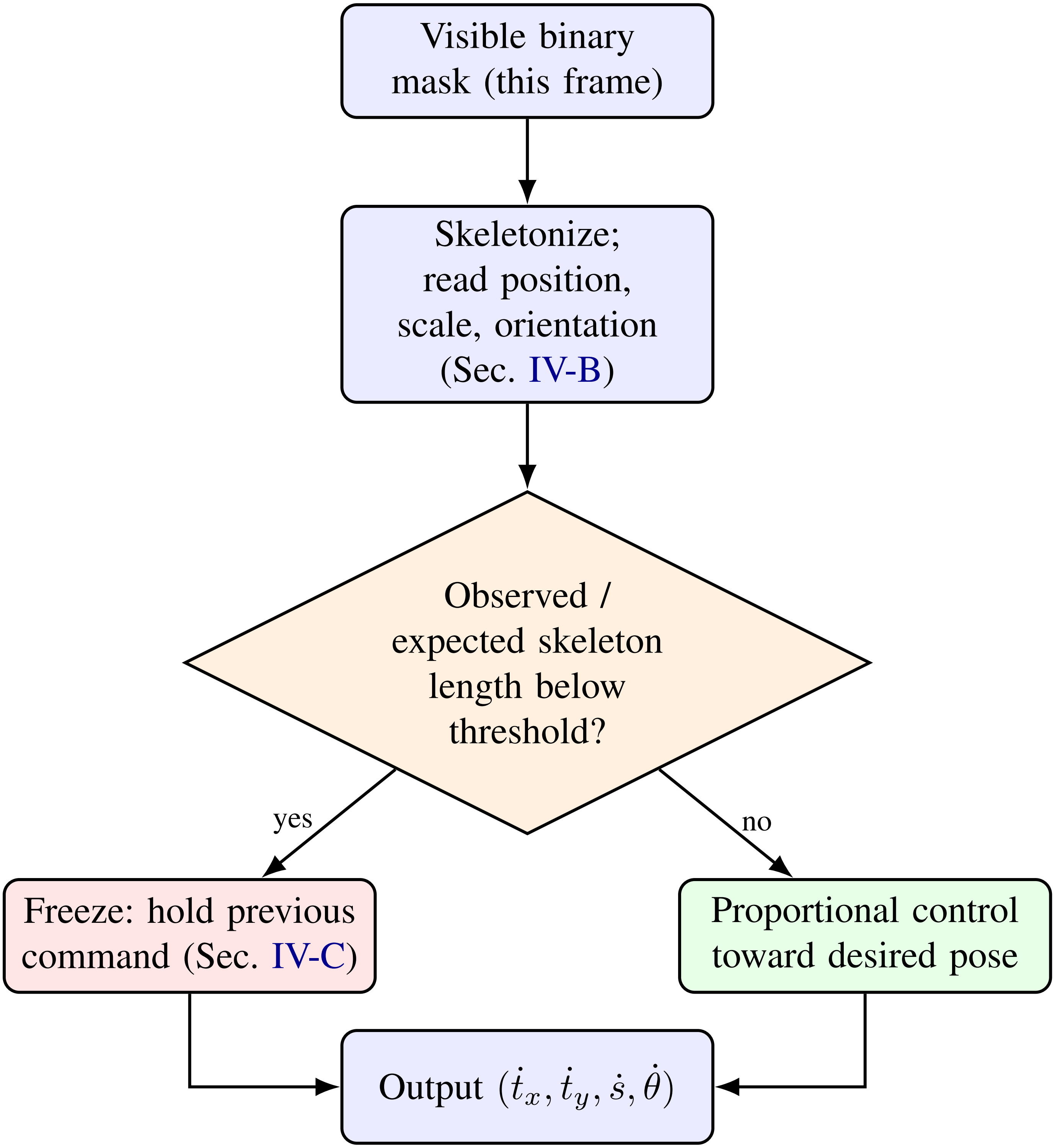

Fig. 3: One SF-AIBVS control cycle. The confidence check compares the skeleton length actually observed to the length a trusted running scale estimate would predict; a large shortfall freezes the output rather than acting on it.

V. Experimental Setup

V-A. Simulated Docking Environment

Each trial renders a T-shaped target silhouette into a 128×128 binary image under a hidden true pose (t_x, t_y, s, θ), initialized with position error up to 45 pixels, scale between 0.4 and 2.0 times the desired value, and rotation up to ± 1.8 radians. A single rectangular occluder sweeps horizontally across the image at a fixed row, sized and

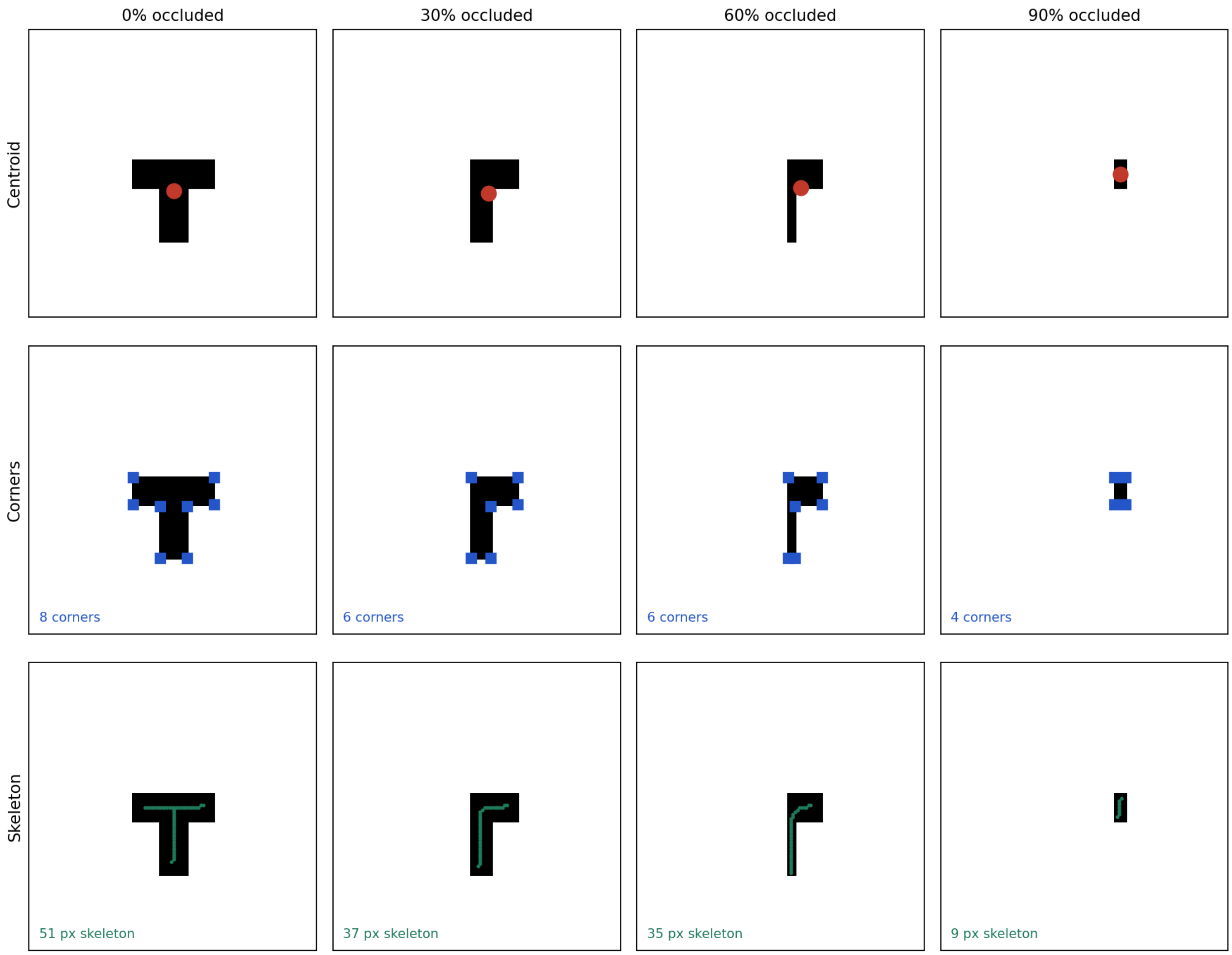

Fig. 4: The identical target rendered at four occlusion levels (0%, 30%, 60%, 90% of silhouette area), read by three feature types. The centroid (top) drifts steadily toward the visible remainder. Corner count and identity (middle) both change unpredictably as area is removed, breaking point correspondence. The skeleton (bottom) shrinks in pixel count but its topology, and the estimate derived from it, degrades continuously rather than discontinuously.

timed according to three severity levels: light (14×14 pixels, engaged with probability 0.5), moderate (26×26, probability 0.8), and heavy (58×58, always engaged, biased toward the row the target typically occupies once converged). Actuation is treated as exact, meaning a controller's commanded change is applied to the hidden pose without additional noise, which isolates differences between controllers to their perception and feature-extraction stage rather than to actuator variability. Thirty random seeds are run per severity per controller, giving 360 trials in total.

V-B. Baselines

Centroid-IBVS computes the mean position and $\sqrt{\text{area}}$ of the visible mask and drives these toward the image center and a target area; it issues no rotational command, since a centroid alone carries no orientation information. **Corner-IBVS** detects corner candidates from the visible mask's contour via polygon approximation, matches them by nearest distance to the positions its current transform estimate predicts, and refits a similarity transform by least squares from whatever correspondences survive; fewer than two survivors leaves the transform, and hence the commanded motion, unchanged for that step. A sanity check on the fitted transform's scale rejects the occasional bad correspondence rather than accepting it outright, in the same spirit as (though far simpler than) consensus-based

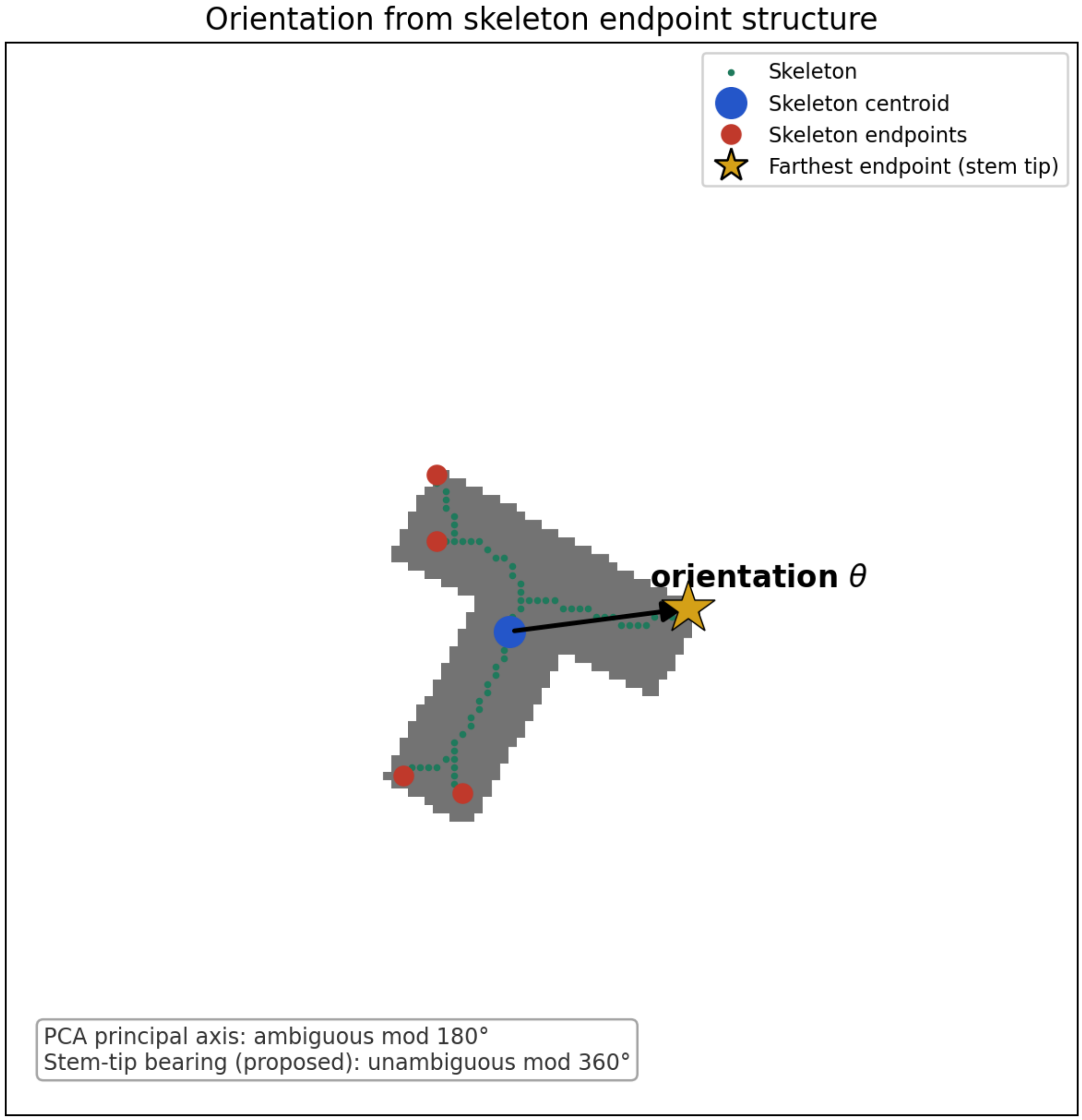

Fig. 5: Orientation from skeleton structure. The bearing from the skeleton centroid (blue) to the farthest endpoint (gold star, among all endpoints marked in red) gives a full 0-to-360-degree orientation estimate, unlike a principal-component axis, which is defined only up to a 180-degree ambiguity and cannot by itself distinguish a shape from its half-turn rotation.

robust fitting [18]. Both baselines received the same continuous-throttling and outlier-rejection treatment during development that SF-AIBVS did, so that comparisons reflect genuine differences between feature types rather than an avoidably weak implementation of either baseline.

V-C. Metrics and Statistical Protocol

Each trial ends in success (pose error under 2 pixels in position, 0.06 in scale, 0.06 radians in orientation, sustained for one step), divergence (position error exceeding 220 pixels or scale leaving $[0.05, 6.0]$), or timeout at 800 steps. We additionally record steps to convergence for successful trials, final pose error components, and a smoothness score, the mean Euclidean distance between consecutive commanded control vectors, as a proxy for how jittery the resulting motion is. Because all controllers are evaluated on the same thirty seeds per severity, comparisons of success are tested with an exact binomial test on discordant pairs, and comparisons of smoothness or steps-

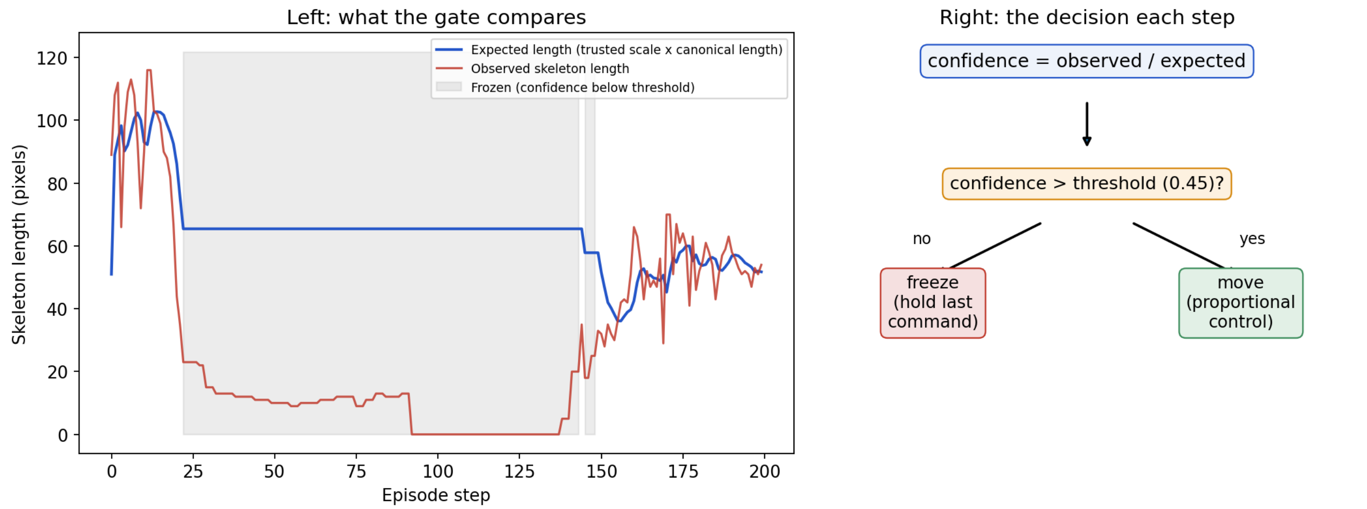

Fig. 6: The gate mechanism on a real heavy-occlusion trial. Left: expected skeleton length (from the trusted scale estimate) against what is actually observed; the shaded region marks steps where the gap is large enough to freeze. Right: the per-step decision the gate makes from that comparison.

Algorithm 1: SF-AIBVS control step

Input: visible binary mask, current trusted scale estimate
Output: control command $(\dot{t}_x, \dot{t}_y, \dot{s}, \dot{\theta})$
skel ← skeletonize(mask);
if *too few skeleton pixels to extract features* **then**
  return $(0, 0, 0, 0)$; `// nothing to measure`
$(t_x^{\text{est}}, t_y^{\text{est}}, s^{\text{est}}, \theta^{\text{est}}) \leftarrow$ extract_features(skel) ; `// Sec. IV-B`
expected_len ← trusted_scale × canonical_skeleton_length;
confidence ← observed_len / expected_len;
if *confidence* $<$ *threshold* **then**
  return $(0, 0, 0, 0)$; `// freeze, Sec. IV-C`
trusted_scale ← (1-α)×trusted_scale + $\alpha \times s^{\text{est}}$;
$\dot{t}_x \leftarrow -k_{\text{pos}}\, t_x^{\text{est}}, \quad \dot{t}_y \leftarrow -k_{\text{pos}}\, t_y^{\text{est}}$;
$\dot{s} \leftarrow -k_{\text{scale}}\, (s^{\text{est}} - s_{\text{desired}})$;
$\dot{\theta} \leftarrow -k_{\text{rot}}\, \theta^{\text{est}}$;
return $(\dot{t}_x, \dot{t}_y, \dot{s}, \dot{\theta})$;

---

to-converge (restricted to trials where both compared controllers succeeded) use the paired Wilcoxon signed-rank test.

VI. Results

VI-A. Success, Divergence, and Timeout

Table I and Fig. 11 summarize outcomes across all four controllers and three severities. SF-AIBVS reached the target in all 30 trials at every severity, including heavy occlusion, where the occluder sometimes covers the entire target for extended stretches (Fig. 2). Centroid-IBVS never once reached the target at any severity; measured at timeout, its orientation error averages 1.03 radians regardless of severity, essentially the same as a random initial rotation, confirming this is a structural limitation of the feature, not a symptom of occlusion, since the controller simply never receives a signal telling it to rotate. Corner-IBVS succeeded in 13.3% to 20.0% of trials and, more tellingly, diverged outright in 13.3% of light trials, 33.3% of moderate trials, and 76.7% of heavy trials, a

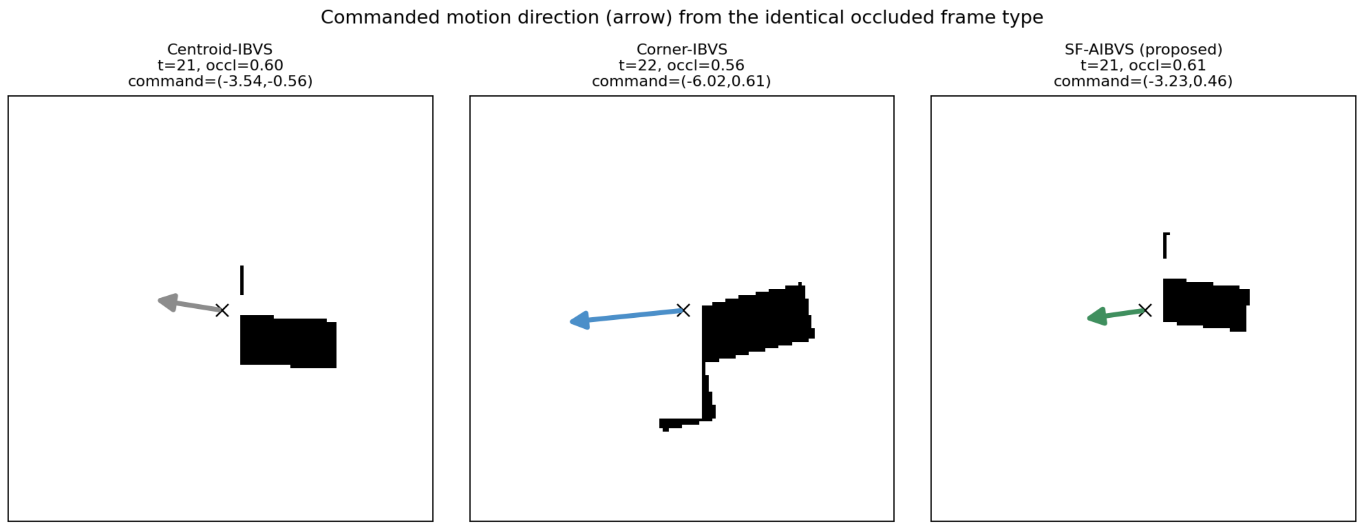

Fig. 7: Commanded motion direction (arrow) from each controller, given the same category of heavily occluded frame (occlusion fraction 0.55-0.75, heavy severity, seed 4). The centroid and corner commands point away from the true correction direction; the skeleton-based command does not.

clean monotonic breakdown as the occluder grows and lingers longer, consistent with the correspondence problem illustrated in Fig. 4.

Fig. 7 makes the consequence concrete: given the identical style of heavily occluded frame, the three controllers do not just disagree on speed, they disagree on direction. The centroid controller's command points toward the visible remainder rather than the true center; the corner controller's fitted transform, thrown off by a changed correspondence set, points in a direction unrelated to the actual pose error; the skeleton-based command still points sensibly toward the goal.

Numbers in a table are one way to see this; watching the three controllers run on the identical trial is another, and the two pictures agree, once read correctly. Fig. 8 shows actual mask frames for all three controllers on the same seed under light occlusion: Centroid-IBVS visibly centers and scales the target correctly but never corrects its rotation and consequently never satisfies the success criterion within the step budget; Corner-IBVS reaches the correct position and scale almost immediately but its fitted rotation estimate locks onto a spurious, self-consistent value and stays there for hundreds of steps before a small perturbation lets it escape and correct within the final dozen steps; SF-AIBVS converges directly. Fig. 9 repeats the comparison under heavy occlusion, a genuinely different regime: there, Corner-IBVS diverges outright partway through as real correspondence loss corrupts its fit, Centroid-IBVS again stalls on rotation, and SF-AIBVS still reaches the goal despite the target being fully hidden at points during the trial. Fig. 10 looks at the light-occlusion trial two ways – as a path in image space, and as orientation error over time – because the position path alone is misleading here: Corner-IBVS's position barely moves after the first 50 steps, so what actually costs it nearly 600 steps is invisible unless orientation is plotted separately.

Table I reports what happened; Table II reports how well the successful trials went. We split these apart deliberately: a single wide table mixing outcome fractions with error magnitudes forces the reader to hold two different kinds of number in mind at once, and the two questions, did it succeed, and when it did, how good was the result, are best read separately.

Corner-IBVS's position error at success, 1.85-1.91 px, sits close to the 2.0 px threshold rather than comfortably inside it, which is consistent with the mechanism Fig. 10 shows directly: a trial only registers as successful the instant its rotation estimate breaks free of the spurious fixed point described above, and that break tends to happen abruptly rather than through gradual, well-damped convergence, leaving less margin on the other two parameters at the moment success is checked than SF-AIBVS has.

Every success-rate comparison between SF-AIBVS and each baseline is significant at every severity ($p < 10^{-6}$ against Centroid, $p < 10^{-6}$ against Corner; Table III), which is unsurprising given how far apart the raw counts

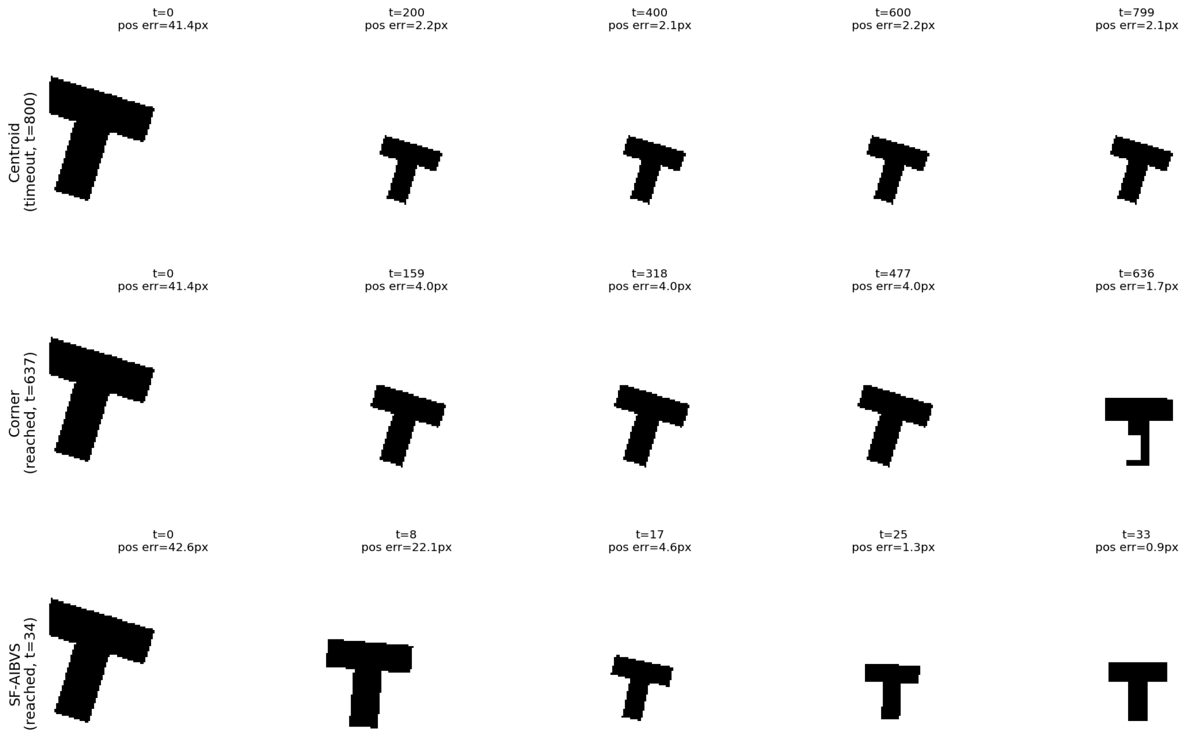

Fig. 8: All three controllers on the identical trial (light occlusion, seed 3), real mask frames at five checkpoints each. Centroid-IBVS (top) centers the target but cannot correct its rotation and times out. Corner-IBVS (middle) reaches the correct position within the first few frames shown here but its rotation estimate is stuck at a wrong, self-consistent value for most of the trial, only correcting in the final steps (see Fig. 10 for the orientation trace this filmstrip cannot show). SF-AIBVS (bottom) converges directly in 34 steps.

TABLE I: Trial outcomes by controller and occlusion severity ($n = 30$ trials each). The three columns sum to 1.000 in every row: every trial ends in exactly one of success, divergence, or timeout.

Severity	Controller	Success	Diverged	Timeout
Light	Centroid	0.000	0.000	1.000
Light	Corner	0.200	0.133	0.667
Light	**SF-AIBVS**	**1.000**	0.000	0.000
Light	SF-AIBVS (no gate)	1.000	0.000	0.000
Moderate	Centroid	0.000	0.000	1.000
Moderate	Corner	0.167	0.333	0.500
Moderate	**SF-AIBVS**	**1.000**	0.000	0.000
Moderate	SF-AIBVS (no gate)	1.000	0.000	0.000
Heavy	Centroid	0.000	0.000	1.000
Heavy	Corner	0.133	0.767	0.100
Heavy	**SF-AIBVS**	**1.000**	0.000	0.000
Heavy	SF-AIBVS (no gate)	1.000	0.000	0.000

are, 30/30 versus 0/30 and 30/30 versus 4-6/30, and reported here mainly for completeness rather than because the reader needs convincing.

VI-B. What the Confidence Gate Actually Changes

Here the honest result is a mixed one. Tables I and II show SF-AIBVS with and without the gate reaching identical success counts at every severity, and at moderate severity the two produce numerically identical steps-to-

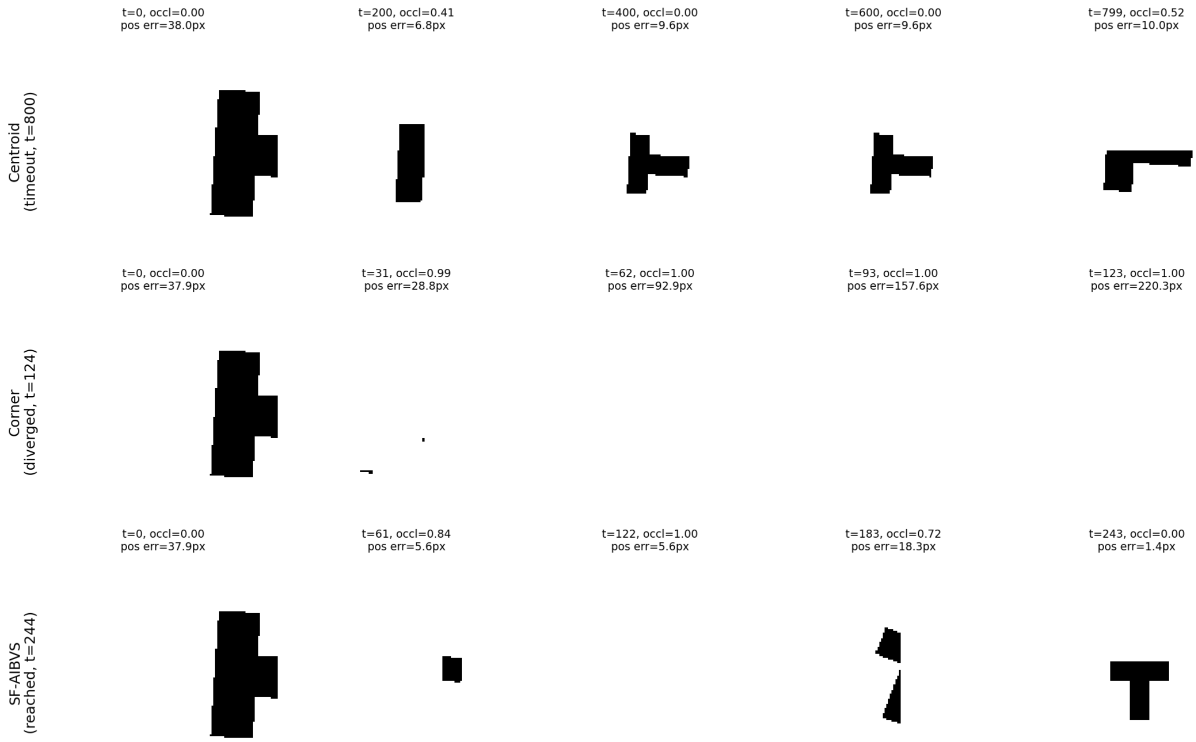

Fig. 9: The same three controllers under heavy occlusion (seed 4). Centroid-IBVS again stalls on rotation. Corner-IBVS diverges outright at step 124 as the occluder disrupts its corner correspondences. SF-AIBVS reaches the goal at step 244 despite the target being completely hidden (occlusion fraction 1.00) at two points during the trial.

TABLE II: Trajectory quality, computed only over trials that succeeded (rows for controllers with zero successes are omitted rather than filled with a meaningless number). Errors are measured at the moment the trial ended.

Severity	Controller	Steps to converge	Pos. err. (px)	Rot. err. (rad)
Light	Corner	122 ± 253	1.85	0.003
Light	**SF-AIBVS**	58 ± 100	0.66	0.028
Light	SF-AIBVS (no gate)	41 ± 16	0.67	0.028
Moderate	Corner	19 ± 7	1.89	0.012
Moderate	**SF-AIBVS**	44 ± 19	0.72	0.028
Moderate	SF-AIBVS (no gate)	44 ± 19	0.72	0.028
Heavy	Corner	19 ± 8	1.91	0.004
Heavy	**SF-AIBVS**	98 ± 91	0.77	0.029
Heavy	SF-AIBVS (no gate)	98 ± 91	0.77	0.026

converge and final errors, meaning the gate simply never triggered in a way that changed the outcome for these particular thirty seeds. The paired Wilcoxon test on steps-to-converge (Table III) confirms no significant timing difference at any severity ($p = 0.44$ light, $p = 0.92$ heavy).

Where the gate does leave a measurable trace is control smoothness under heavy occlusion. Fig. 12 and Table IV show the gated controller producing a mean command-to-command change of 1.08 under heavy occlusion versus 1.29 without the gate, a 19% reduction that is statistically significant ($p = 0.0069$, paired Wilcoxon, $n = 30$). At light and moderate severity the difference is not significant ($p = 0.32$ and undefined, respectively, since every command was identical in the moderate case). Fig. 13 shows why: in a representative heavy-occlusion trial the gate is frozen for 126 of 244 steps, more than half the trial, actively suppressing the jittery corrections that raw, partially-occluded skeleton readings would otherwise produce, even though the eventual outcome (success, and in roughly the same number of steps) would have been the same either way.

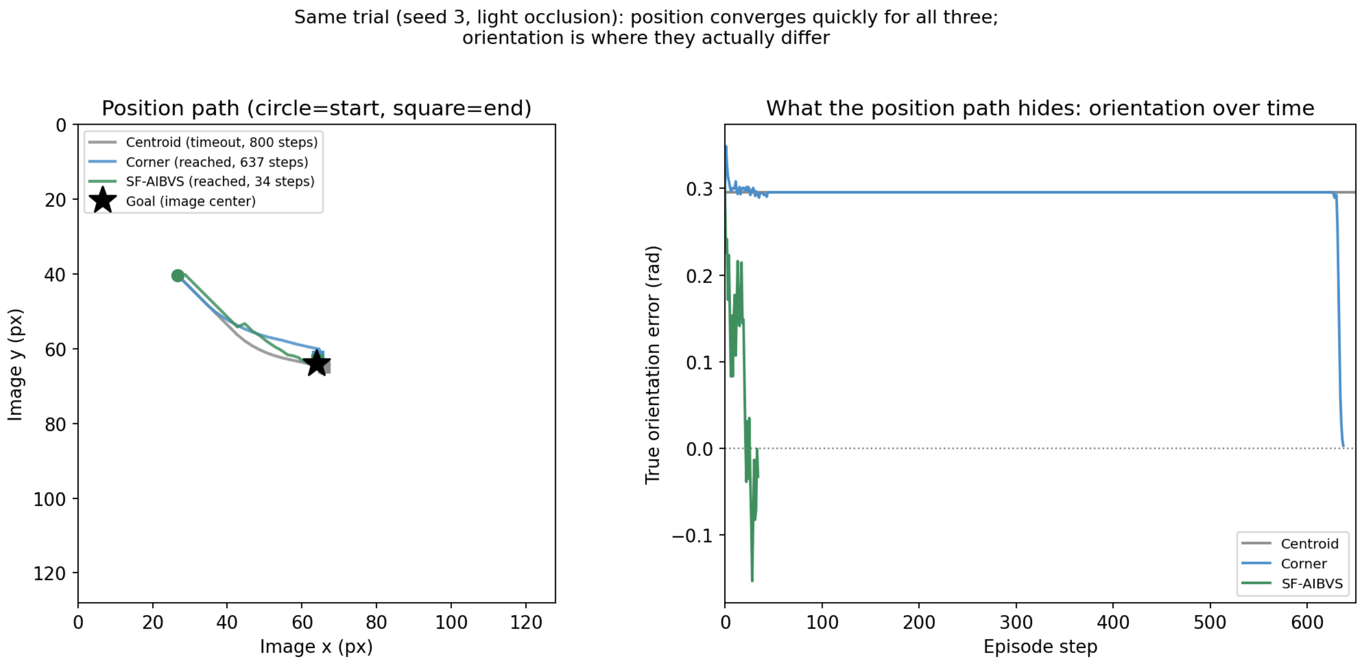

Fig. 10: The same light-occlusion trial (seed 3), read two ways. Left: each controller's path in image space (circles mark the start, squares the end) – SF-AIBVS moves directly to the goal, Centroid-IBVS settles near it in position but never in orientation, and Corner-IBVS's path looks similarly direct, which is the trap. Right: true orientation error over time, which the position path cannot show. Corner-IBVS's rotation estimate locks onto a wrong value within the first 50 steps and sits there, essentially unchanged, until step 626, then corrects within the final 11 steps. Its position path looks clean only because the robot is otherwise nearly motionless for the intervening 575 steps.

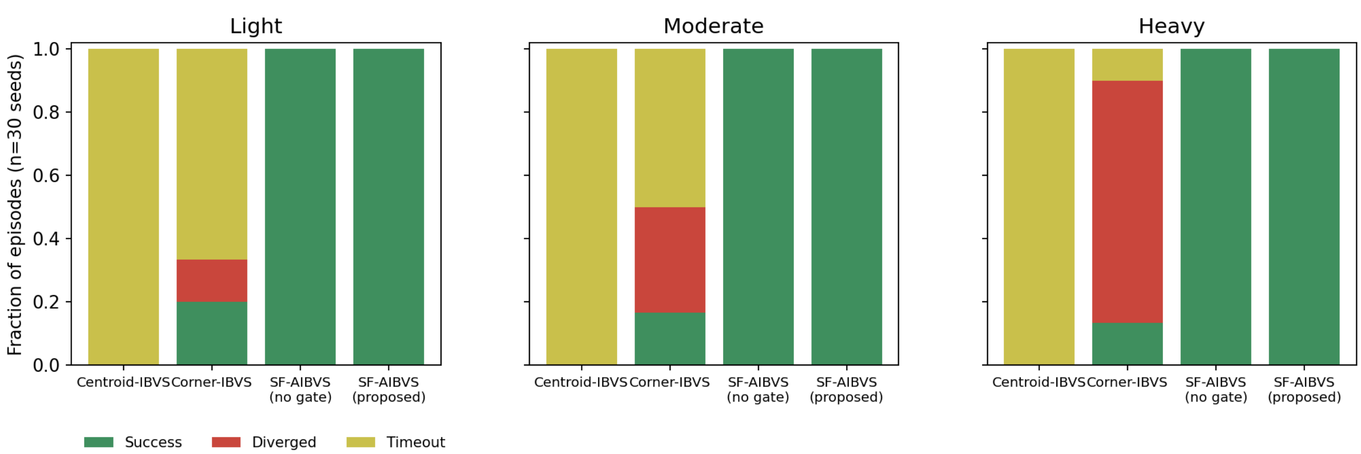

Fig. 11: Success (green), divergence (red), and timeout (yellow) fractions by controller and occlusion severity, $n = 30$ trials each.

VI-C. Computational Cost

SF-AIBVS runs at roughly 0.42 to 0.46 ms per control step, dominated by the skeletonization call; Centroid-IBVS is cheapest at about 0.05 ms and Corner-IBVS's contour and polygon-approximation calls cost around 0.11 to 0.12 ms. All three remain far inside any reasonable real-time control budget for a docking task of this kind. We have not profiled how skeletonization cost scales with image resolution or measured a GPU implementation; the numbers above are single-threaded CPU timings at 128×128 resolution only, and should not be extrapolated to, for instance, a 1080p feed without remeasuring.

TABLE III: Paired significance tests, SF-AIBVS versus each comparison, by severity ($n = 30$; exact binomial test on discordant success outcomes; Wilcoxon signed-rank on steps-to-converge, restricted to trials where both compared controllers succeeded).

Severity	Comparison	SF-AIBVS	Other	Success p	Steps p
Light	vs. Centroid	1.000	0.000	< 0.0001	—
Light	vs. Corner	1.000	0.200	< 0.0001	0.4375
Light	vs. no gate	1.000	1.000	1.0000	0.3173
Moderate	vs. Centroid	1.000	0.000	< 0.0001	—
Moderate	vs. Corner	1.000	0.167	< 0.0001	0.0625
Moderate	vs. no gate	1.000	1.000	1.0000	—
Heavy	vs. Centroid	1.000	0.000	< 0.0001	—
Heavy	vs. Corner	1.000	0.133	< 0.0001	—
Heavy	vs. no gate	1.000	1.000	1.0000	0.9188

TABLE IV: Control smoothness (mean per-step command change; lower is smoother) with and without the confidence gate.

Severity	With gate	Without gate	Wilcoxon p
Light	0.847 ± 0.232	0.865 ± 0.189	0.317
Moderate	0.911 ± 0.230	0.911 ± 0.230	—
Heavy	1.082 ± 0.441	1.287 ± 0.654	**0.0069**

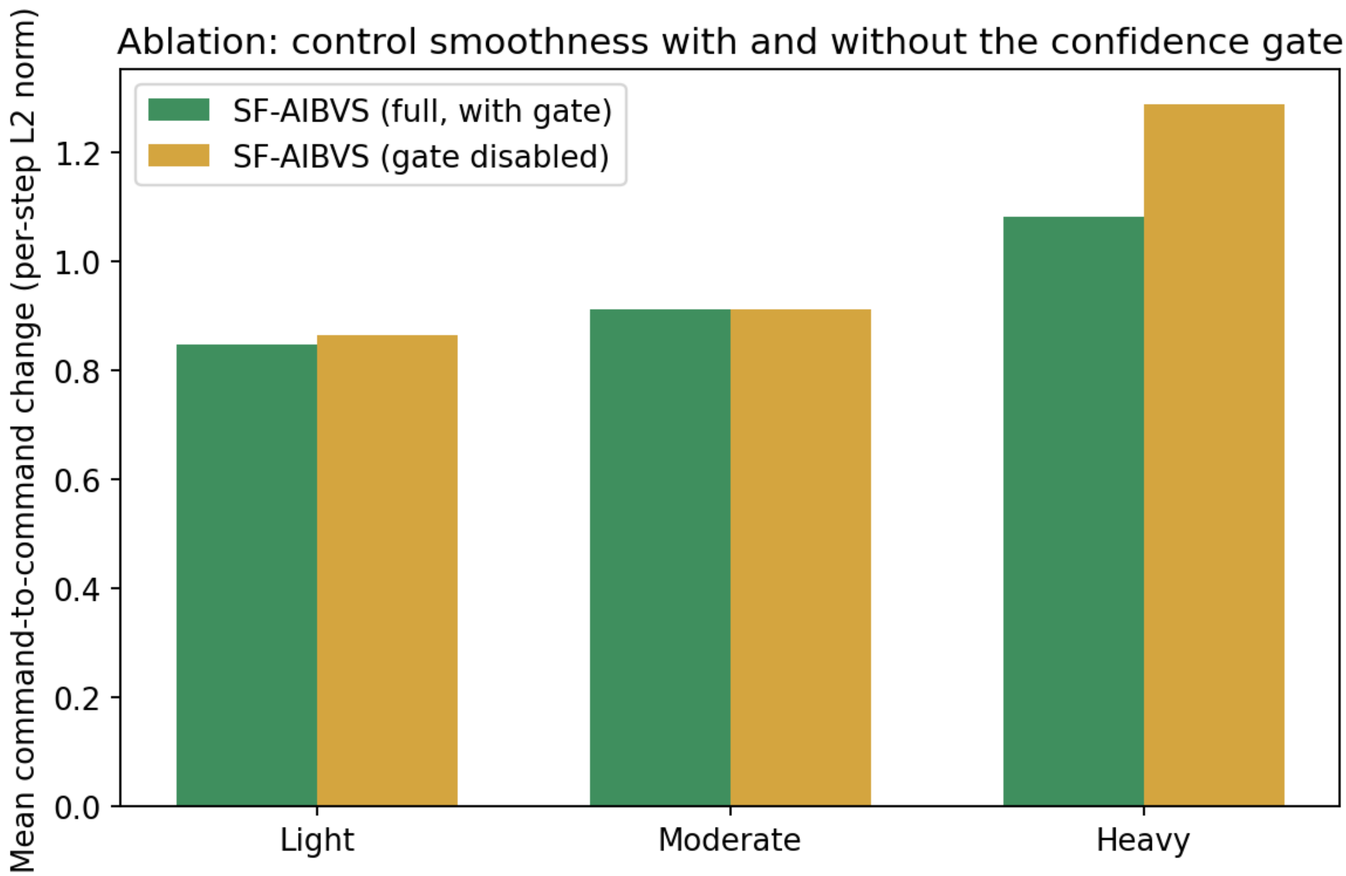

Fig. 12: Control smoothness with and without the confidence gate. The gap widens with occlusion severity and is significant only at the heavy setting.

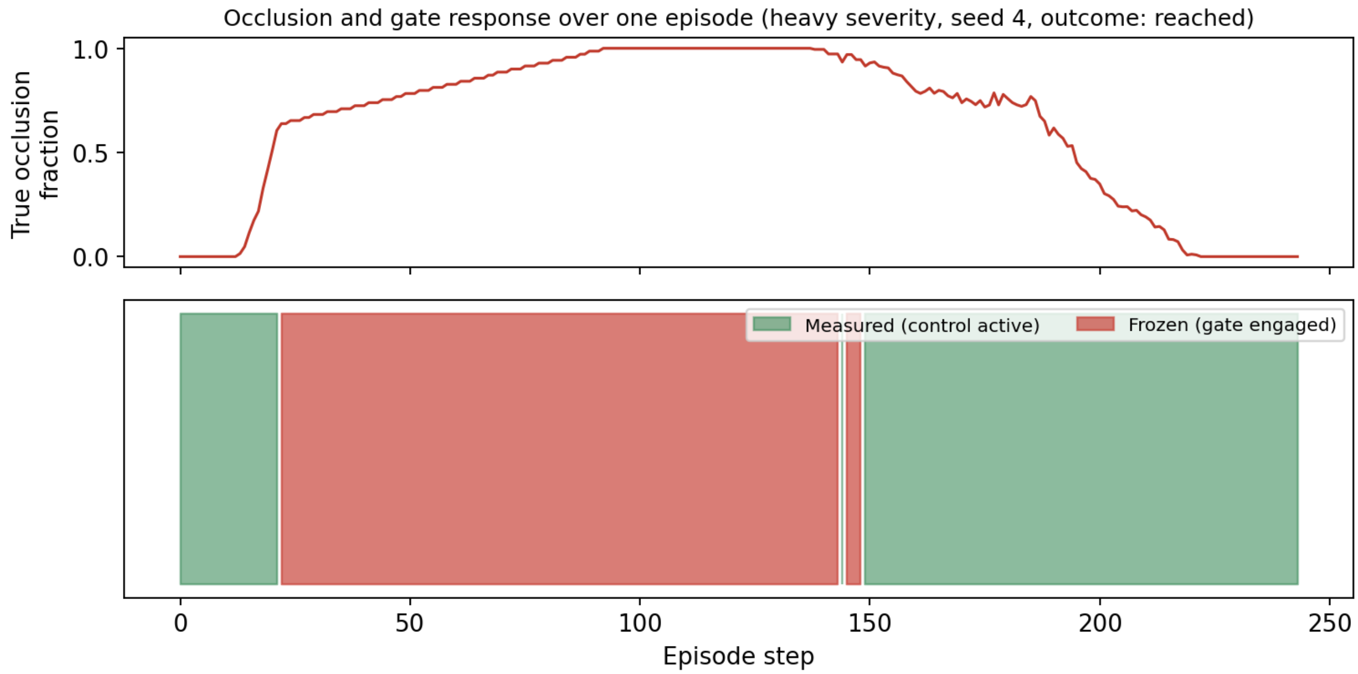

Fig. 13: True occlusion fraction (top) and gate state (bottom) across one heavy-occlusion trial. The gate freezes output during the two near-total-occlusion intervals visible in the top panel, and control resumes within a few steps of the target becoming visible again.

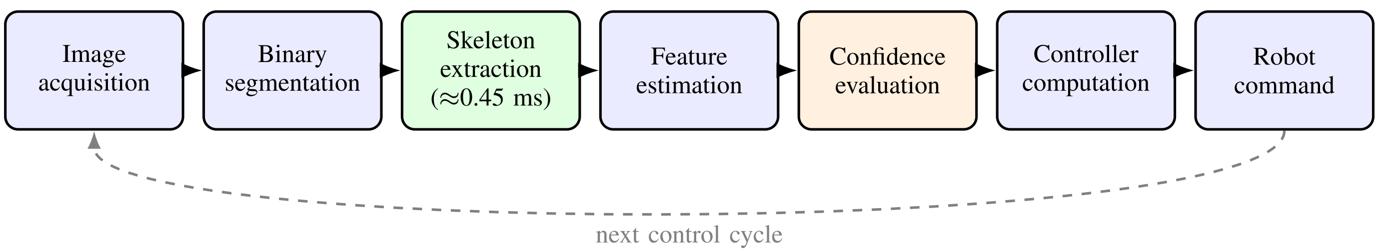

Fig. 14: One SF-AIBVS control cycle. Skeleton extraction is the single most expensive stage, at roughly 0.45 ms, but the full cycle still completes in under half a millisecond, comfortably inside any typical real-time control budget.

VI-D. Sensitivity to the Confidence Threshold

The results above use a fixed confidence threshold of 0.45, chosen during development and not tuned against the 360-trial evaluation set. Because a fixed threshold picked without justification is a reasonable thing to be skeptical of, we swept it from 0.30 to 0.90 and reran all three severities at each value (Fig. 15). Success rate is flat at 100% for thresholds at or below 0.45, then degrades once the threshold exceeds roughly 0.5: 22–28 of 30 trials still succeed at 0.60, but only 9–14 of 30 at 0.75, and success collapses almost entirely by 0.90 (0–1 of 30). The failure mode at high thresholds is not divergence, it is timeout: an overly strict threshold rejects perfectly usable measurements as unreliable often enough that the controller spends most of an episode frozen and never accumulates enough progress within the step budget. This tells us the chosen value of 0.45 sits inside a wide, flat plateau rather than at a knife-edge, which is some reassurance against having tuned it luckily, but it is not the same as a principled derivation of the right threshold from first principles, and we do not claim one.

VII. Discussion

VII-A. The Feature Choice Is Doing Most of the Work

The clearest finding in this paper is not about the confidence gate at all: it is that a skeleton-based feature set, on its own, tolerates occlusion that breaks centroid and corner features outright. That result held even when we

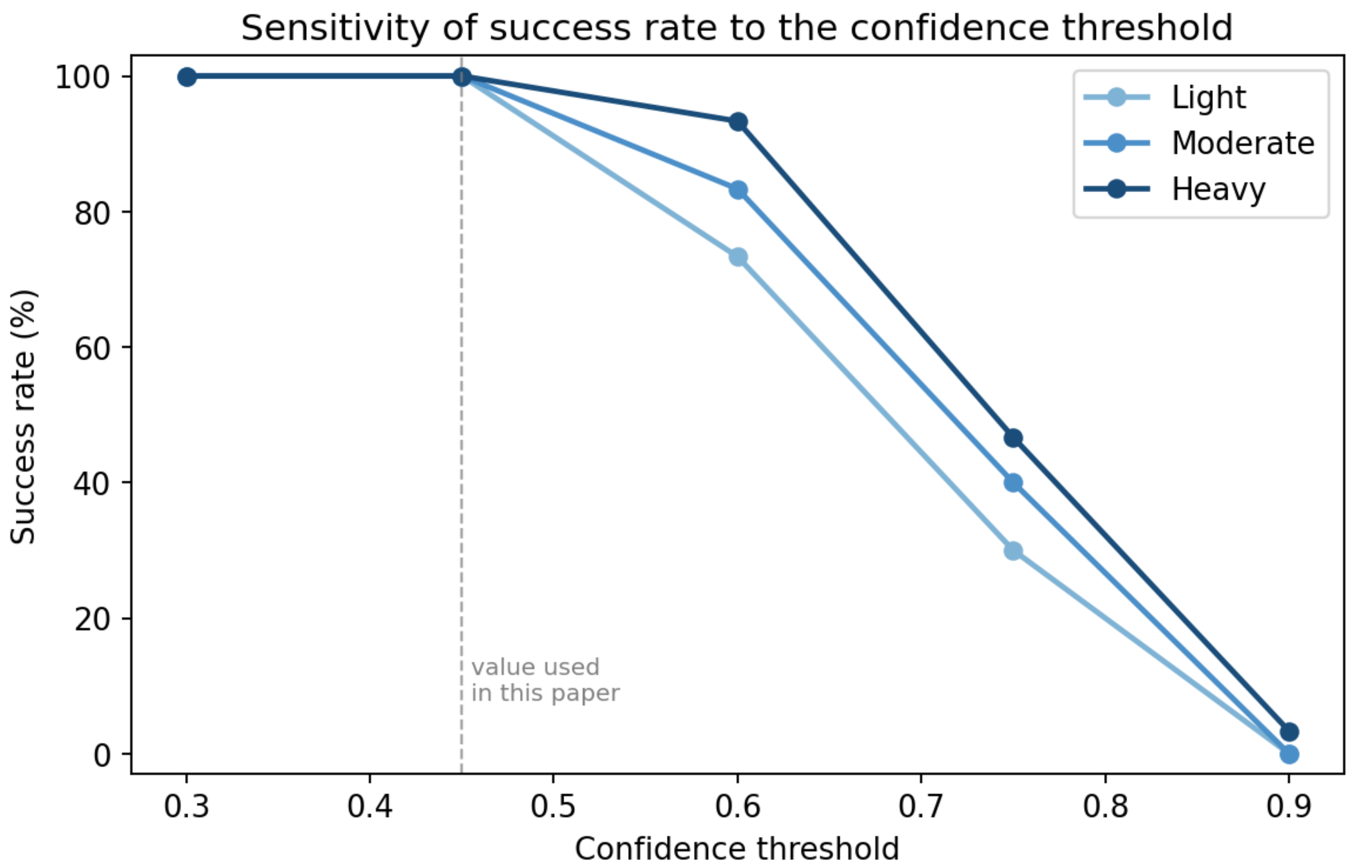

Fig. 15: Success rate as a function of the confidence threshold, all three severities. The value used throughout this paper (0.45, dashed line) sits well inside a plateau where performance is insensitive to the exact choice; performance degrades once the threshold is raised past roughly 0.5, and collapses by 0.9 because the controller freezes too readily even on clean measurements.

deliberately pushed occlusion to an extreme, a 58-pixel occluder crawling across the image at a third of its normal speed, covering the target almost continuously, and it held whether or not the gate was active. We think this is worth stating plainly rather than folding into a more impressive-sounding combined claim: the gate is a secondary refinement, not the reason SF-AIBVS works.

VII-B. Why We Chose to Freeze Rather Than Predict

It is worth recording, briefly, a design path that did not survive contact with testing. Our first confidence-gated controller extrapolated the position, scale, and orientation estimate forward using its recent rate of change whenever confidence dropped, in the spirit of the predict step of a recursive filter [17], on the reasonable-sounding assumption that a target moving at roughly constant velocity should keep being trackable through a brief occlusion. In practice, once the controller had converged and the target's true velocity was near zero, any residual extrapolated velocity, even a small one caused by ordinary measurement noise just before the occlusion began, got applied every single step for as long as the occlusion lasted. Over a hundred-plus-step heavy occlusion this produced a slow, then accelerating, drift that occasionally carried the estimated pose (and, through the proportional controller, the true pose) far enough off course to diverge outright, despite the target never having actually moved. Freezing sidesteps this specific failure by refusing to issue any command, correct or not, while confidence is low; the cost is that a frozen controller makes zero progress during occlusion instead of possibly making the right kind of progress. Given that our experiments show the raw, ungated skeleton readings were already reliable enough that this trade rarely matters for the final outcome, freezing was the more defensible choice for this system, though a more sophisticated bounded-drift predictor is a reasonable direction for a target that genuinely keeps moving during long occlusions

(Section VIII).

VII-C. Threats to Validity

We would rather list these plainly than have a reader discover them unaided.

Simulation-only validation. This is the largest gap in the paper, and we do not think it can be argued away. Every result here comes from synthetic binary masks, exact actuation, and an orthographic camera model; real segmentation, whether a classical threshold-based method [19], an edge-based one [20], or a learned instance segmentation network [21], produces jagged, occasionally broken silhouette boundaries that our clean rasterized masks never do, and skeletonization is well known to be sensitive to exactly this kind of boundary noise. We have not measured what happens to the stem-tip orientation estimate (Sec. IV-B) when the skeleton itself has spurious short branches from boundary noise, and we would not be surprised if it requires a pruning step (removing branches below a minimum length before endpoint selection) that the current implementation lacks entirely. Until this is tested on real camera images, the results in this paper should be read as evidence that the skeletal-feature idea is worth testing on hardware, not as evidence that it will survive the transition unmodified.

One target shape. Every trial uses the same T-shaped silhouette. The stem-tip orientation estimate (Sec. IV-B) works because the T has one branch clearly longer than the others; a shape with two branches of similar length (a plus sign, a symmetric cross), or one with rotational symmetry (a hexagonal nut, a circular valve handle), would leave the farthest-endpoint rule ambiguous or, for a symmetric shape, would make orientation fundamentally unobservable from silhouette alone regardless of feature choice. Tool-like shapes with a genuine handle, a hammer, wrench, or screwdriver, seem like the natural next test, since they share the T's single-dominant-branch structure; shapes without that structure, such as a mug or gear, would likely need a different orientation cue (skeleton branch topology and relative branch lengths, rather than the single farthest point) and are not covered by anything in this paper.

Reduced pose model. The controller estimates four parameters, image-plane translation, isotropic scale, and in-plane rotation, not a full six-degree-of-freedom pose under perspective projection. Extending the skeletal feature idea to out-of-plane rotation and a perspective interaction matrix is a real piece of additional work, not a minor adjustment, since skeleton branch length under perspective foreshortening behaves differently than it does under the orthographic scaling assumed here.

Baseline strength. We compare against centroid and corner-based IBVS, both long-established but not the only reasonable choices; image moment features, keypoint descriptors such as SIFT [14], SURF [15], or ORB [16], fiducial-marker servoing with AprilTag [22] or ArUco [23], and optical-flow-based tracking in the tradition of Lucas-Kanade [24] are all plausible alternatives we did not implement. We chose centroid and corner features because they isolate two specific, well-understood failure modes, no rotation information and correspondence loss, that motivate the skeletal approach directly, but a reader should not conclude from this paper alone that skeletal features outperform every reasonable alternative; only that they outperform these two particular ones, for the reasons given.

Perfect success may reflect an easy task, not a strong method. A 100% success rate across every severity we tested is exactly the kind of result that should prompt suspicion rather than satisfaction, and we do not think the concern is misplaced. Part of the answer is visible in this paper's own data: the same task, same occlusion sequences in several cases, drove one baseline to 0% and another to 13–20% with frequent divergence (Table I), so the task is not trivial in any absolute sense. But we have not added sensor noise, control latency, camera motion, or realistic segmentation error, any of which could plausibly take SF-AIBVS off its current ceiling, and until those are added we cannot rule out that the simulation is easier for this specific method than a real deployment would be.

Statistical power. Thirty simulated trials per condition, seeded independently, is a defensible sample size for the paired statistical tests we report, but 360 total simulated trials are not equivalent to 360 independent physical experiments; every trial shares the same simulator, the same actuation model, and the same rendering of occlusion, so systematic errors in any of those (rather than trial-to-trial noise) would affect all 360 trials in the same direction and would not be caught by more seeds alone.

No stability or convergence guarantee. The controller is entirely empirical. We do not provide a Lyapunov argument, a proof of local or global convergence, or a formal characterization of the basin of attraction the proportional control law operates within, particularly given that the confidence gate makes the closed-loop system switched and discontinuous (control authority changes abruptly at the confidence threshold) rather than a plain continuous feedback loop, a structure with its own dedicated stability theory [25] that this paper does not draw on. This is a legitimate gap for readers who need a certified controller rather than an empirically validated one.

Threshold and gain sensitivity. Sec. VI-D shows the confidence threshold sits inside a wide plateau rather than at a narrow optimum, which is reassuring, but the position, scale, and rotation gains (Algorithm 1) were not swept the same way and were fixed during development on a small number of exploratory runs, not tuned on data separate from the 360 trials reported in Sec. VI.

VIII. Conclusion and Future Work

Skeletal features, used directly as the state a visual servoing controller acts on, tolerate occlusion far better than centroid or corner features do: 100% success across 90 trials spanning three occlusion severities, against 0% for a centroid controller and 13-20% for a corner-based one, with the corner controller's divergence rate climbing to 77% under heavy occlusion, and even its successes typically arriving after an order of magnitude more steps than the proposed method needs (Fig. 8). A confidence gate built on the ratio of observed to expected skeleton length adds a measurable, statistically significant reduction in control jitter under heavy occlusion, without changing the success rate, and a sweep of its threshold from 0.30 to 0.90 shows the value used throughout this paper sits inside a wide plateau rather than a tuned knife-edge. We have tried to report all of this, including the parts that limit rather than support the method, as plainly as the parts that support it.

That plainness extends to what this paper has not done, and Section VII-C lists it in full; the following is a priority order for closing those gaps, not a repeat of the list. The single most important next step is validation against a real camera and a real segmentation stage, since every other result in this paper is downstream of the assumption that the visible mask is clean, and that assumption is the one most likely to be wrong in practice; a pruning step that removes short spurious skeleton branches before endpoint selection is a plausible and inexpensive place to start hardening the feature extractor for that transition. Second, testing tool-like shapes with a genuine dominant handle (a hammer, wrench, or screwdriver) would show whether the stem-tip orientation idea generalizes at all beyond the single T-shape tested here, before attempting the harder case of shapes without a dominant branch. Third, adding realistic stressors, sensor noise, control latency, camera motion, and comparing against stronger and more varied baselines (image moments, keypoint-based servoing, fiducial markers, learned features) together would test whether the 100% success rate reported here is a genuine property of the method or an artifact of an unrealistically forgiving simulation, which is a question this paper's data cannot settle on its own. Fourth, a formal stability or convergence analysis of the switched system the confidence gate creates would move this from an empirically validated controller to a certified one, which matters for any deployment context that requires more than empirical confidence. Extending the feature set to full six-degree-of-freedom pose recovery and replacing the freeze-on-low-confidence gate with a bounded-drift predictor for targets that keep moving during long occlusions, designed deliberately to avoid the unbounded-extrapolation failure described in Section VII rather than reintroducing it, round out the list.

Acknowledgment

The authors thank the Department of Computer Science and Information Engineering, Asia University, for supporting this research.

References

[1] B. Espiau, F. Chaumette, and P. Rives, "A new approach to visual servoing in robotics," *IEEE Transactions on Robotics and Automation*, vol. 8, no. 3, pp. 313–326, 1992.

[2] S. Hutchinson, G. D. Hager, and P. I. Corke, "A tutorial on visual servo control," *IEEE Transactions on Robotics and Automation*, vol. 12, no. 5, pp. 651–670, 1996.

[3] H. Blum, "A transformation for extracting new descriptors of shape," in *Models for the Perception of Speech and Visual Form*, W. Wathen-Dunn, Ed. Cambridge, MA: MIT Press, 1967, pp. 362–380.

[4] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. Cambridge, MA, USA: MIT Press, 2005.

[5] T. Y. Zhang and C. Y. Suen, "A fast parallel algorithm for thinning digital patterns," *Communications of the ACM*, vol. 27, no. 3, pp. 236–239, 1984.

[6] A. C. Sanderson and L. E. Weiss, "Adaptive visual servo control of robots," in *Robot Vision*, A. Pugh, Ed. Bedford, U.K.: IFS Publications, 1983, pp. 107–116.

[7] F. Chaumette and S. Hutchinson, "Visual servo control, Part I: Basic approaches," *IEEE Robotics & Automation Magazine*, vol. 13, no. 4, pp. 82–90, 2006.

[8] F. Chaumette and S. Hutchinson, “Visual servo control, Part II: Advanced approaches,” *IEEE Robotics & Automation Magazine*, vol. 14, no. 1, pp. 109–118, 2007.
[9] D. Comaniciu, V. Ramesh, and P. Meer, “Kernel-based object tracking,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 25, no. 5, pp. 564–577, 2003.
[10] Q. Bateux, E. Marchand, J. Leitner, F. Chaumette, and P. Corke, “Training deep neural networks for visual servoing,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 3307–3314.
[11] Y. F. Chen, M. Liu, M. Everett, and J. P. How, “Decentralized non-communicating multiagent collision avoidance with deep reinforcement learning,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2017, pp. 285–292.
[12] C. Harris and M. Stephens, “A combined corner and edge detector,” in *Proc. 4th Alvey Vision Conference*, Manchester, U.K., 1988, pp. 147–151.
[13] J. Shi and C. Tomasi, “Good features to track,” in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1994, pp. 593–600.
[14] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.
[15] H. Bay, T. Tuytelaars, and L. Van Gool, “SURF: Speeded up robust features,” in *Proc. European Conference on Computer Vision (ECCV)*, 2006, pp. 404–417.
[16] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, “ORB: An efficient alternative to SIFT or SURF,” in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 2564–2571.
[17] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *Transactions of the ASME – Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.
[18] M. A. Fischler and R. C. Bolles, “Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography,” *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981.
[19] N. Otsu, “A threshold selection method from gray-level histograms,” *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.
[20] J. Canny, “A computational approach to edge detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 8, no. 6, pp. 679–698, 1986.
[21] K. He, G. Gkioxari, P. Dollár, and R. Girshick, “Mask R-CNN,” in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2961–2969.
[22] E. Olson, “AprilTag: A robust and flexible visual fiducial system,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2011, pp. 3400–3407.
[23] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez, “Automatic generation and detection of highly reliable fiducial markers under occlusion,” *Pattern Recognition*, vol. 47, no. 6, pp. 2280–2292, 2014.
[24] B. D. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” in *Proc. 7th International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.
[25] D. Liberzon, *Switching in Systems and Control*. Boston, MA, USA: Birkhäuser, 2003.